

"Osprey 3.5 V6: Harsh shifts and stored fault from failing shift solenoid pack"

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1 Condition

Some 3.5L V6 Osprey vehicles may exhibit harsh or delayed upshifts, flaring, or a default-to-limp condition accompanied by DTC P0700 (Transmission Control System Malfunction). The malfunction indicator lamp may illuminate when the transmission control module requests an MIL request through P0700. Drivers often report erratic shift quality that worsens as fluid temperature rises. See Osprey Transmission Service Manual :: Diagnostic Trouble Code Reference for the codes that accompany P0700.

2 Cause

DTC P0700 on these vehicles is most often caused by a failing shift solenoid pack TSOL-9520 that produces electrical or hydraulic shift errors, which the control module reports as a system malfunction. Contaminated fluid or a worn valve body can contribute to the condition. Review transmission operation in Osprey Transmission Service Manual :: Transmission System Overview and rule out converter-related complaints using Osprey Transmission Service Manual :: Torque Converter Clutch Diagnosis before condemning TSOL-9520.

3 Diagnosis

Perform transmission diagnostic routine PROC-TRANS-DIAG to read the underlying component codes stored alongside P0700, since P0700 itself is a generic flag set by the TCM. Verify fluid level and condition, then command each solenoid and monitor pressure response to isolate the faulty circuit. If the data confirms a solenoid electrical or pressure fault, the shift solenoid pack TSOL-9520 is the confirmed failure. The complete sequence is documented in Osprey Transmission Service Manual :: Transmission Fault Diagnosis Procedure .

4 Repair Procedure

Replace the shift solenoid pack TSOL-9520 following procedure PROC-SHIFT-SOL, taking care to keep the valve body clean and to torque fasteners to specification. After installation, perform the adaptive relearn, verify shift quality, and confirm DTC P0700 does not return after a road test. Replace the transmission fluid and filter if contamination is found. Detailed steps appear in Osprey Transmission Service Manual :: Shift Solenoid Pack Replacement Procedure , with related work grouped under Osprey Transmission Service Manual :: Shift Control Sub-System .